SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I G.Yeshwanth.Ram 192311397 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

G. Yeshwanth Ram

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2. Formulating the 8-puzzle as a Search Problem:

a) State Representation:

- A 3×3 grid containing tiles numbered 1-8 and one blankspace.

b) Critical State:

- The starting arrangement of the tiles provided before solving the puzzle.

Example:

2	8	3
1	6	4
7	_	5

c) Goal test

- The puzzle is solved when the tiles are arranged in the desired order.

Example Goal state:

1	2	3
4	5	6
7	8	_

d) Possible Actions:

The blank tile can move: - Up, down ~~and~~, left and Right.

Why is branching factor at most 4?

- The blank tile has at most four neighboring position (up, down, left, right)

- At corners, it has only 2 possible moves.
- Along edges, it has only 3 possible moves.
- At the center, it has only 4 possible moves.

∴ the maximum branching factor is 4.

PEAS Description and Environment of a Tic-Tac-Toe Agent:

Component	Description
Performance Measure	Win the game, avoid losing, maximize draws if winning is impossible
Environment	3x3 Tic Tac Toe board and opponent
Actuators	Place an X or O in a empty cell
Sensors	Observe all the current board configuration and opponents' moves

Environmental Characteristics:

- fully Observable: Yes both player can see the entire board at all times
- Deterministic: Yes. A move always produces a predictable result.
- Episode or Sequential. Sequential, because each move affects future moves.
- Single agent or multiagent Since two players interact.
- Static or Dynamic: Static, because the board changes only when a player makes a move.
- Discrete or Continuous: Discrete, Since the board has a finite number of states and actions.

Why is it Adversarial?

The environment is adversarial because there are two competing agents with opposite goals. Each player tries to maximize their own chance of winning while preventing the opponent from winning. Every move made by one player directly influences the other player's strategy, making it a competitive environment.

CO, Assessment tool A.I

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CSA 17

1. Warehouse Robot Agent Architecture.

Most appropriate architecture : Utility based architecture

A utility based agent is the most suitable architecture for warehouse robots because it not only works towards achieving a goal but also evaluates different possible actions and selects that one that provides the highest overall benefit. In a warehouse, robots must transport goods efficiently while avoiding obstacles, minimizing travel time, conserving battery power and ensuring safety.

Reasons:

- The Robot can choose the shortest and least congested path to the packing station
- It considers multiple factors such as travel time, battery level traffic and other robots and obstacle avoidance before making a decision.
- It can avoid collision with workers shelves, and other moving robots, improving workplace safety.
- If several routes are available, it selects the one with the highest utility balancing efficiency and safety.
- It increases overall warehouse productivity by completing tasks faster while reducing energy consumption and delays.

Conclusion:

A utility-based agent is ideal because warehouse operation require intelligent decision-making that optimizes efficiency, safety and resource utilization rather than simply reacting to the environment.